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UNIT - IIIrd

Syllabus :-

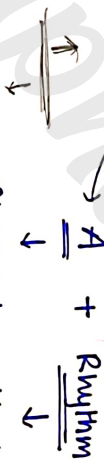
- ① Anti-arrhythmic drugs
 - ② Anti-hyperlipidemic agents
 - ③ Coagulants & Anticoagulants
 - ④ Drug used in congestive heart failure (CHF)
- } Part 1
- Most imp drugs & their synthesis -
 - ① Disopyramide phosphate
 - ② Warfarin

CHAPTER - 1st

ANTI-ARRHYTHMIC DRUGS

These are those drugs which are used in the treatment of arrhythmia or cardiac arrhythmia.

Arrhythmia



- Abnormal/Irregular heart beat
- It is the improper beating of the heart, whether irregular (too fast & too slow).
- It occurs -
 - when electric impulses in the heart does not work properly
 - Also occurs due to other heart disease such as myocardial ischemic disease etc.

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- Normal range of heart beat $\rightarrow$ 60-100 bpm. normally about 72 bpm.

Arrhythmia

Bradycardia

- When heart beat is below 60 bpm.

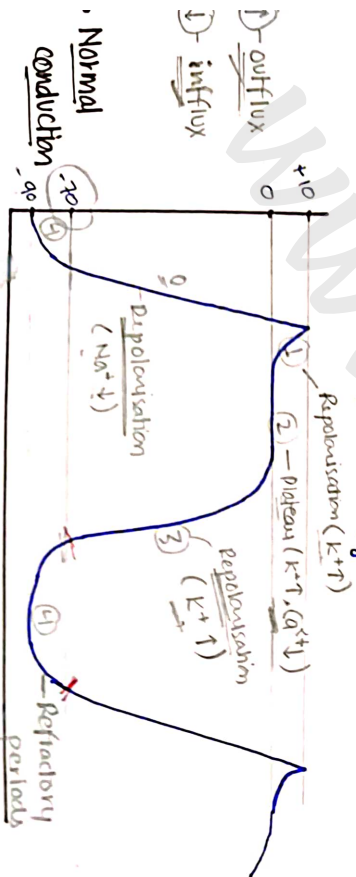
Tachycardia

- When heart beat is more than 100 bpm.

Bradycardia $< 60 - 100 <$ Tachycardia

- Must watch, conduction system of heart and electrophysiology of heart.

Refer Ch-1, Unit-1, pharmacology 5th sem.



- Causes $\rightarrow$ Arrhythmia

Bradycardia

- Reduce Automaticity
 - $\rightarrow$ Parasympathetic system.

— sleeping

— Disease (Heart disease)

— other drugs such

as beta blockers,

antiarrhythmic drugs etc.

- Conduction block

- $\uparrow$ Refractory period

- Some important types $\rightarrow$

- Supraventricular arrhythmia

$\hookrightarrow$ Supra (above) + ventricular

Tachycardia

- Increase automaticity

— $\uparrow$ Sympathetic system.

— Exercise $\uparrow$

- Triggered activity $\uparrow$

early delayed

After depolarisation (EAD) (DAD)

- $\uparrow$ Action potential

- Drugs ($\uparrow$ conduction)

- Re-entry

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When arrhythmia begins in atrium.

i) Atrial flutter (AfI) -

A condition in which our heart's upper chamber (atria) beat too quickly about 400-350 bpm.

ii) Atrial fibrillation (AfF) -

An irregular, mostly rapid heart beat about 350-550 bpm.

It is due to abnormal twitching of heart muscles.

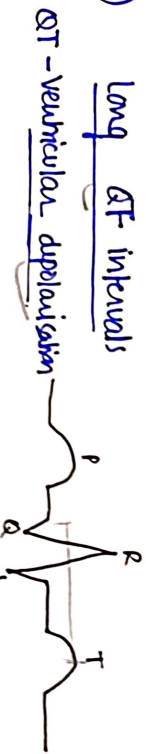
1) Ventricular arrhythmia

Arrhythmia begins in ventricles.

i) Ventricular fibrillation (VF) →

A life threatening heart rhythm that results in a rapid, inadequate heartbeat.

ii) Long QT intervals



• Classification of Antiarrhythmic drugs

- Given by Vaughan Williams and Singh (1969).

① Class-I - Also known as membrane stabilizing agents and Na⁺ channel blockers.

- further divided into three parts :-

i) I-A → eg. Quinidine sulphate, Procainamide HCl.

ii) I-B → eg. Disopyramide phosphate, Pravastatin

iii) I-C → Lorcainide HCl, sodium, lidocaine hydrochloride, Tocainide HCl, Mexiletine HCl.

② Class-II - Also known as Beta blockers.

eg. Propranolol, esmolol, Atenolol etc..

③ Class-III - Potassium Channel blockers.

eg. flecainide, sotalol etc..

④ Class-IV - Calcium channel blockers. (non-DHPs)

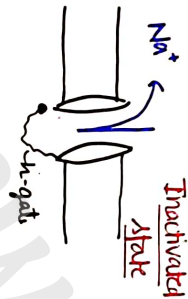
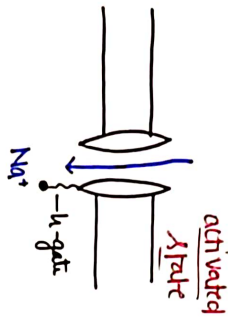
eg. Verapamil, Diltiazem etc..

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① Class - I → Also known as a Sodium Channel blockers.

• Drugs of this class is interfere with Na^+ channel and inhibit the influx of Na^+ , which is responsible for depolarisation.



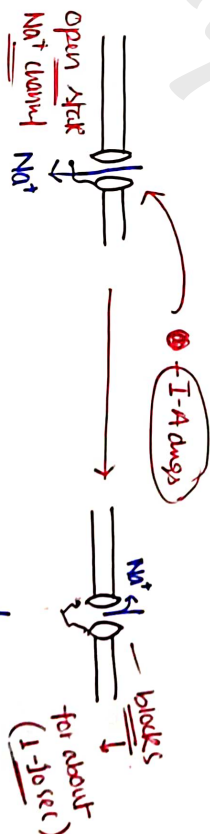
- Now, Due to inhibition of Na^+ channel, concentration of Na^+ in is decreased inside the cell which causes relaxation (repolarisation).
- Which further decreases the depolarisation of phase 0, which gives time to our heart to relax or normalise.
- Divided into three sub parts:—

• Class I-A ⇒

— These drugs blocks the open state Na^+ channel

— Which also delayed the channel recovery time from 1-10 sec.

— Due to this, these drugs decrease conduction & increase refractory periods - decrease Automaticity.



Relaxation of Heart → ↓ Depolarisation

Drugs → • Quinidine Sulphate

• Procainamide hydrochloride

• Disopyramide phosphate

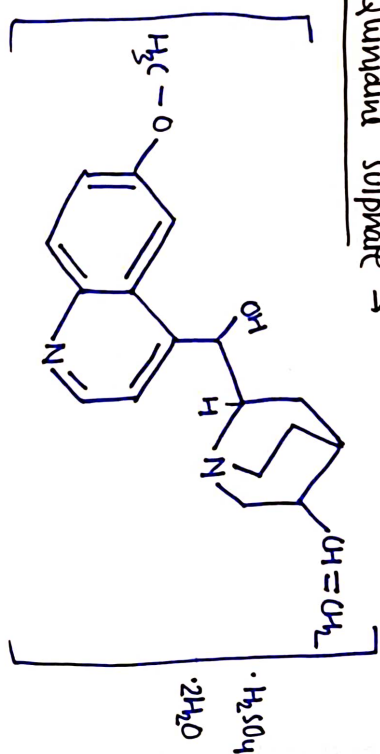
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① Quinidine sulphate →

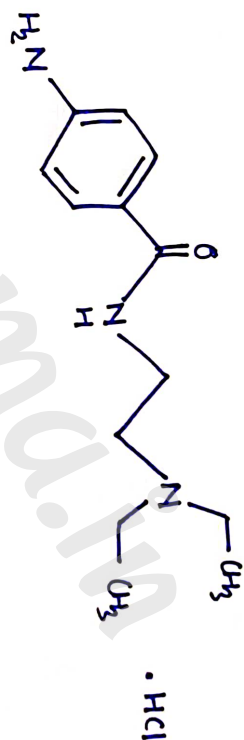


It is a cinchona alkaloids obtain from the bark of cinchona.

uses →

- used in the treatment of supraventricular and ventricular arrhythmias.
- sometimes used to treat malaria as a alternate of quinine.
- also used to treat atrial flutter and atrial fibrillation.
- Also used to treat or abolish re-entry.

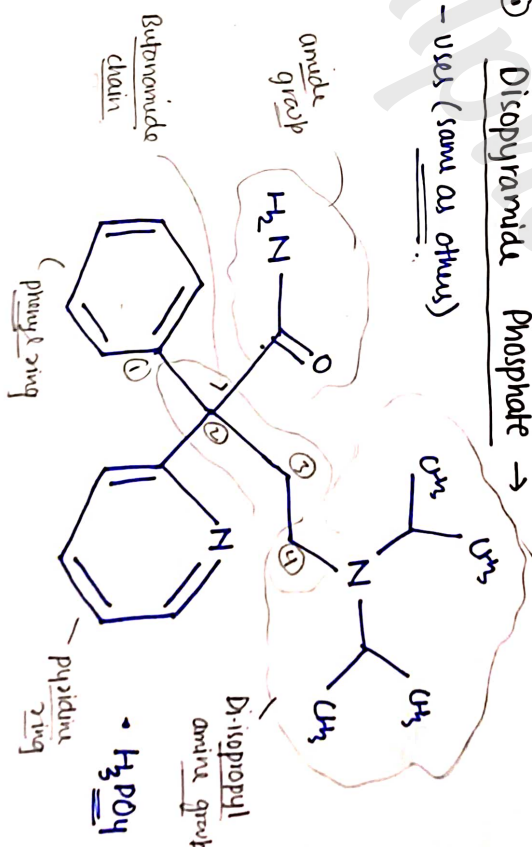
① Procainamide hydrochloride →



— same as Quinidine sulphate (uses / moa).

① Disopyramide Phosphate →

— uses (same as others)



SAR — Structure activity Relationship

- ① Phenyl ring is essential for activity, substitution in ring or removal may reduce

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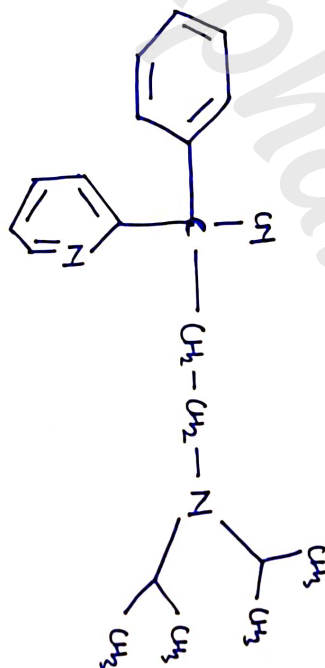
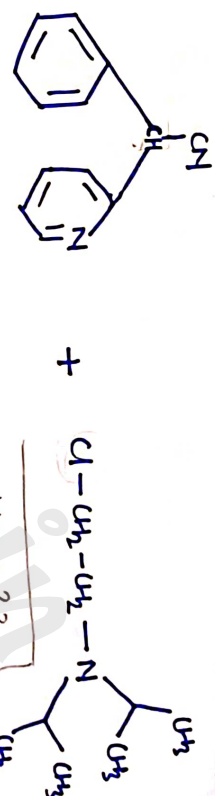
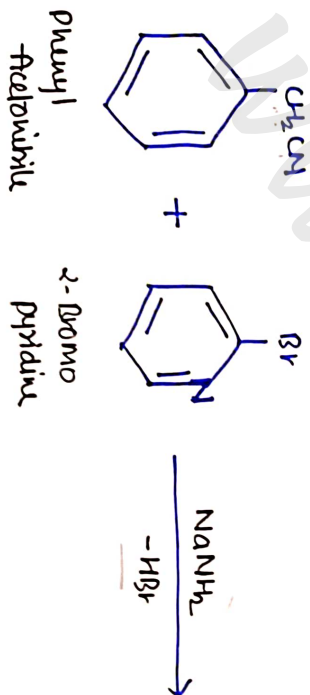
its activity.

③ Pyridine ring at position 2nd is essential for activity.

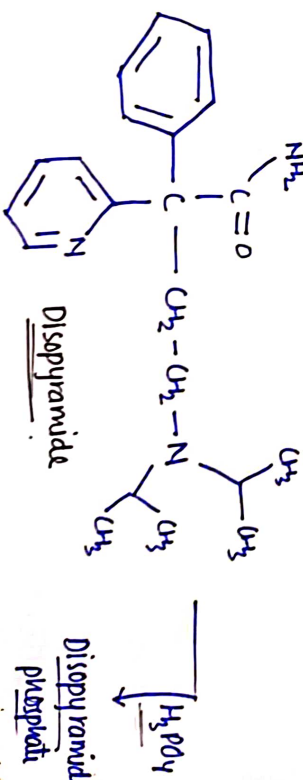
③ Di-isopropylamino group is essential at position 4th, it may be replaced with some small lipophilic structure but it may decrease its activity.

④ amide group is also essential for activity
- It may be replaced with secondary amides or Nitriles but it reduces activity but it gives some anticonvulsant activity.

⑤ Synthesis →



Partial hydrolysis
 $\xrightarrow{\text{H}_2\text{SO}_4}$



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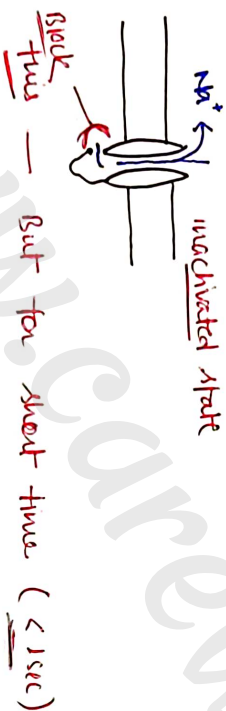
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• class I-B $\Rightarrow$ (weak / moderate)

MOA

- These drugs blocks inactivated state Na^+ channel
- But these drugs not delayed channel recovery time ($< 1\text{s}$).
- These drugs works for very short time
- Decrease automaticity, triggered activity

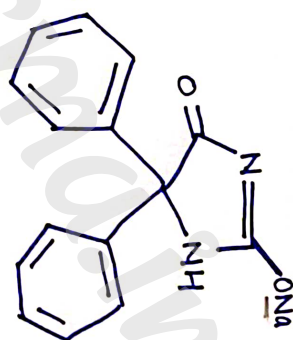


Drugs $\rightarrow$

- Phenytoin Sodium
- Lidocaine hydrochloride
- Tocainide hydrochloride
- Mexiletine hydrochloride

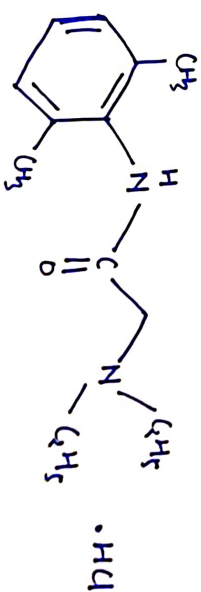
• Phenytoin Sodium $\rightarrow$

- Uses $\rightarrow$
- primarily used as an anti-epileptic in treatment of various types of seizures.



- used in the treatment of cardiac arrhythmia.

• Lidocaine hydrochloride $\rightarrow$



• Uses $\rightarrow$

- used as local anesthetic
- also used in the treatment of arrhythmia.

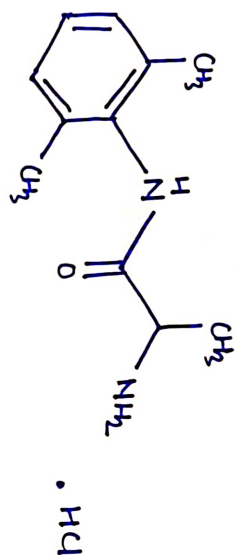
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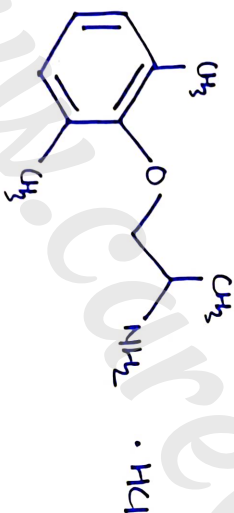
① Tocainide hydrochloride →



uses -

- used as local anesthetic.
- used as anti-arrhythmic

② Mexiletine hydrochloride →



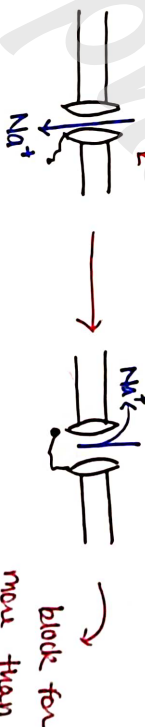
uses -

- used in the treatment of ventricular arrhythmia.
- also have some local anesthetic action and some anticonvulsant action.

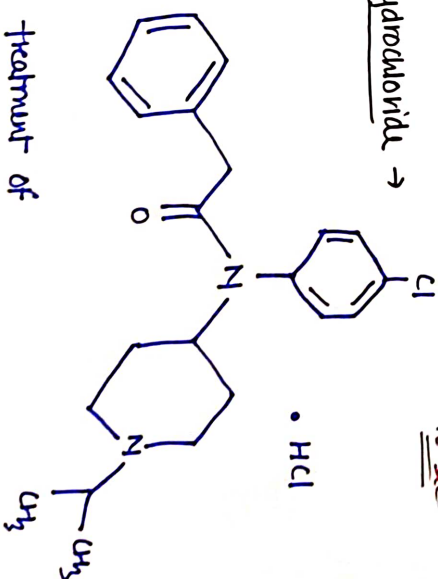
③ Class I-C ≠ (most potent)

NOT

- These drugs block open state Na^+ channel
- But these drugs have longest channel recovery time (more than 10 sec).
- So, it is most potent Na^+ channel blocker.
- Decrease conduction ($\downarrow$ automaticity).



④ Lorcainide hydrochloride →



uses -

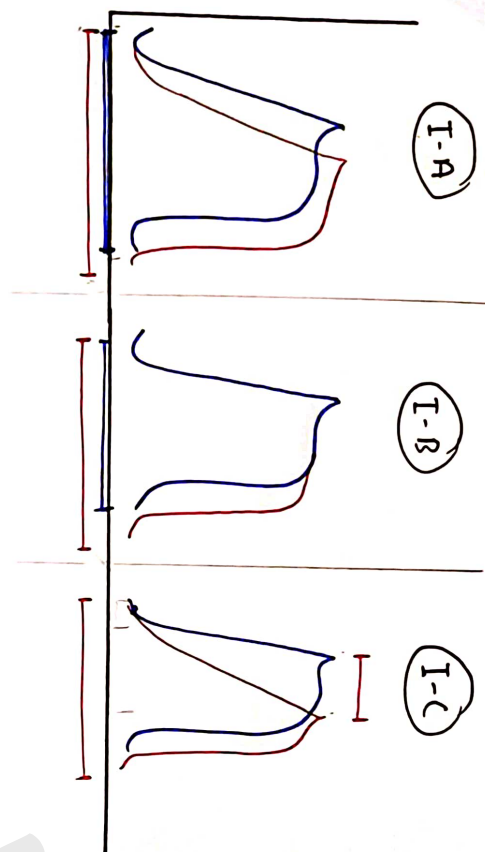
- used in the treatment of ventricular and supraventricular arrhythmia.
- helps to restore the normal heart conduction and rhythm.

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• Class I MOA Graph

②

Class - III →

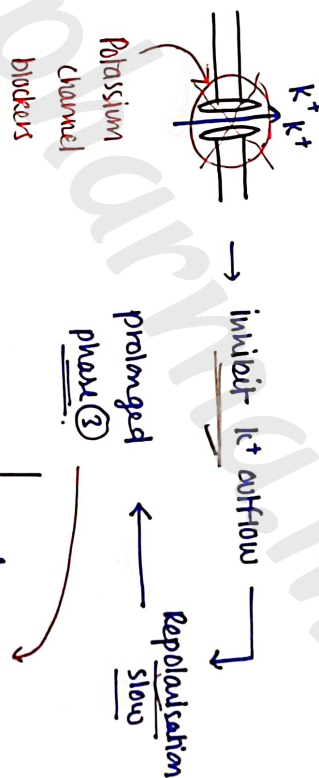
Also known as Potassium channel

blockers.

- These are those drugs which block the potassium channel, which inhibit the outflow of K^+ ion.

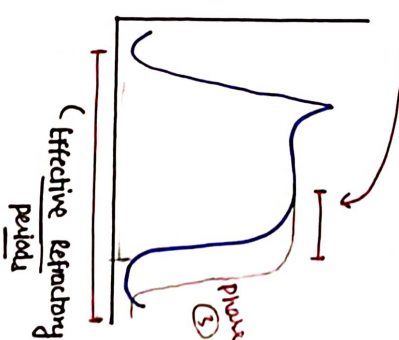
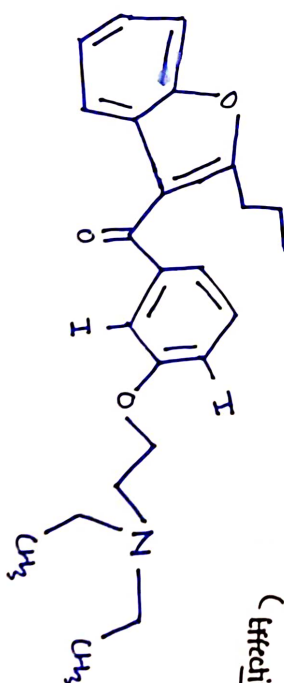
- Due to this, phase 3 (repolarisation) is prolonged ~~the~~ ^{and} duration of action potential is also prolonged.

- It also prolong the effective refractory period
- Also prolong the QT interval.



① Drugs -

① Amiodarone → 3^{rd}



Uses -

- useful in the treatment of ventricular and supraventricular arrhythmia.
- also have some vasodilation properties.

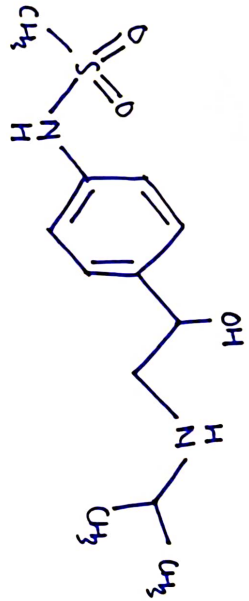
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⊙ Sotalol →



Uses -

- It also have some beta blocker activity.
- used in the treatment of documented life - threatening ventricular arrhythmias.
- also used as in the treatment of cardiac arrhythmia, hypertension, angina pectoris and myocardial infarction.
- re-entry, atrial flutter.

CHAPTER - IInd

ANTI-HYPERLIPIDEMIC AGENTS

- These are drugs which are used in the treatment of hyperlipidemia.
- Also known as -
- lipid lowering agents
- Hypolipidemic drugs

HYPERLIPIDEMIC

→ • hyper (High)
+
• lipidemic (level of lipids)

- hyperlipidemia is a condition in which plasma level of lipid is elevated (increase) in the form of lipoprotein.
- Now, this hyperlipidemia cause atherosclerotic which further cause heart-attack, stroke and many other heart disease.

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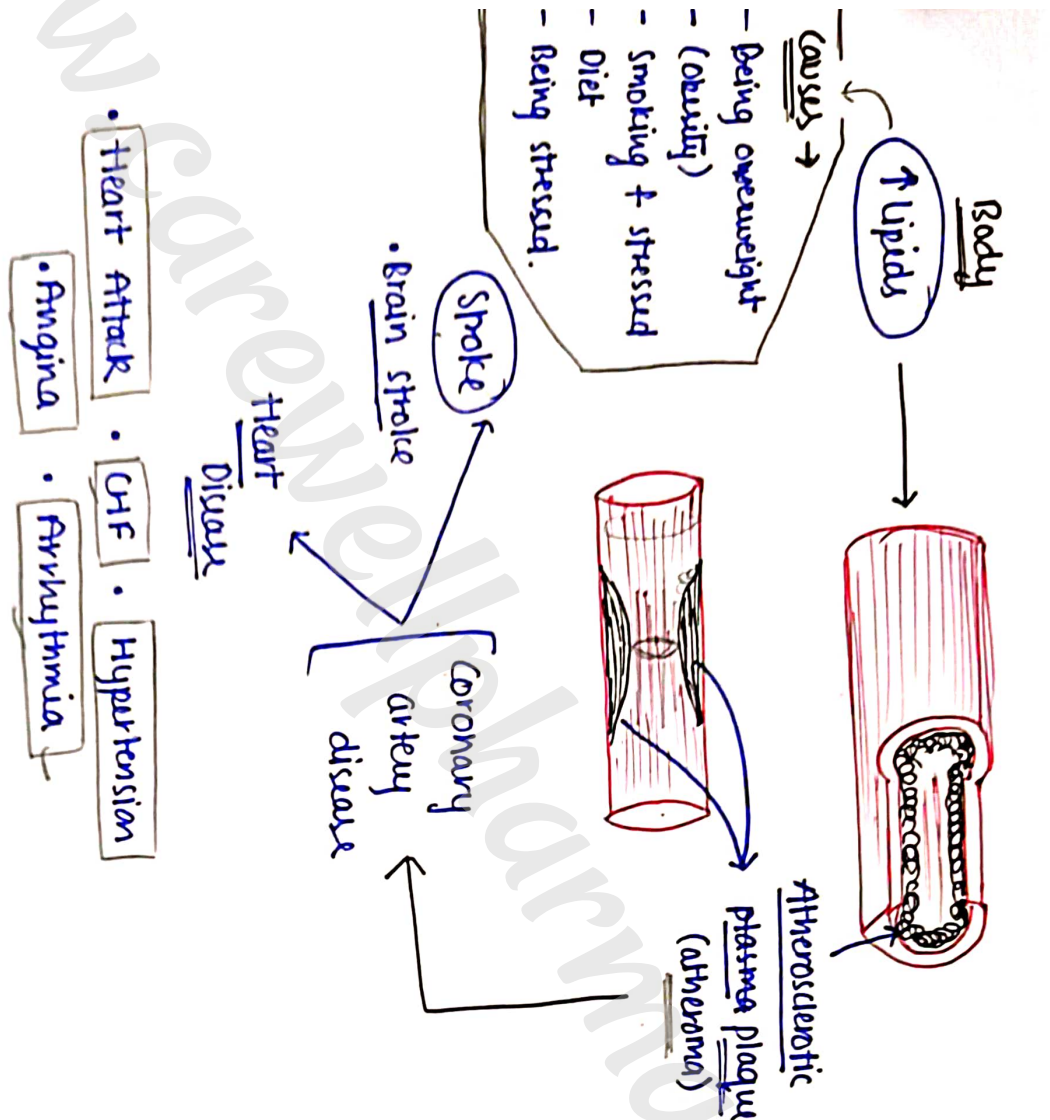
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① Lipids →

These are important for our body.
Now these are present in our body in three forms:—



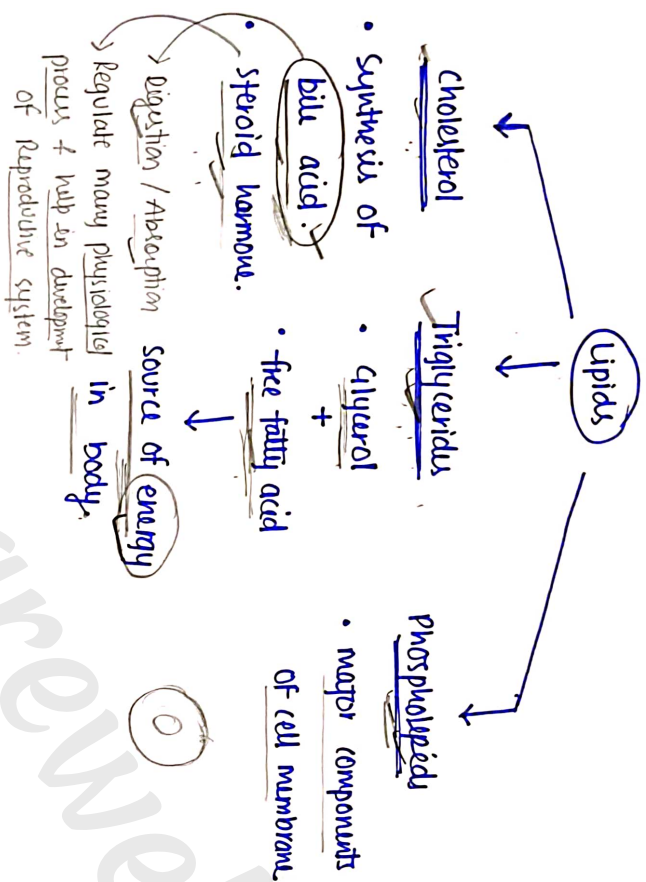
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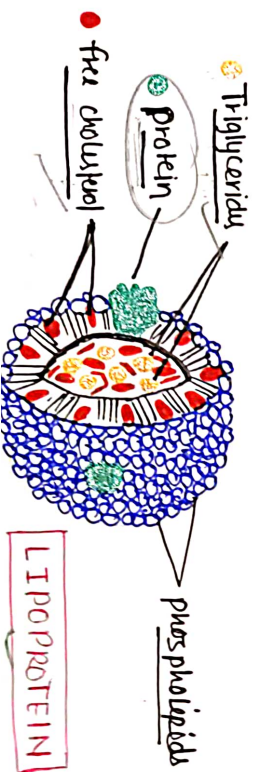
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- Lipid present in three forms in our body:-



- Now, these lipids are insoluble in blood plasma so they have to be transported throughout the body in the form of protein complex which is known as lipoproteins.



- Lipoprotein → cholesterol + triglycerides

lipid substance

which bind non-covalently with proteins.

- Now, These lipoproteins further divided into four major types:-

- i) chylomicrons
- ii) Very low Density lipoproteins (VLDL)
- iii) Low Density lipoproteins (LDL)
- iv) High Density lipoproteins (HDL)

- i) Chylomicrons →

- produced in gut (Gastrointestinal tract) from dietary lipids and small amount of cholesterol.
- It composed of mostly triglycerides and cholesterol.
- functions → Deliver energy rich triglycerides to cells throughout the body.

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ii) VLDL - Very low Density Lipoprotein →

- Produced in liver and gut
- It composed primarily of triglycerides and small amount of cholesterol (more than chylomicrons).

• function → Deliver energy rich triglycerides to body

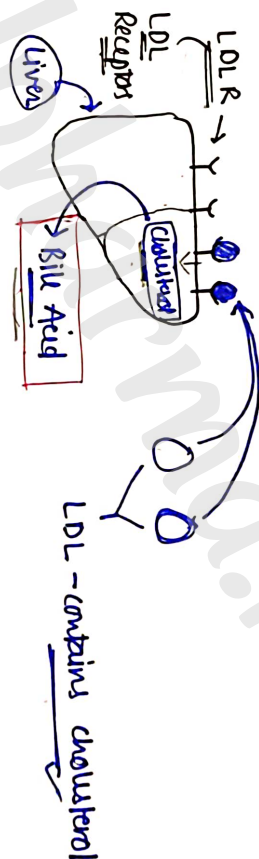
Now, VLDL secreted into blood stream that time triglycerides (fatty acid) are removed from VLDL by the enzyme lipoprotein lipase which is located on the capillary walls and formed LDL (low density lipoproteins).

iii) LDL - Low density Lipoprotein →

- produced from VLDL
- contain higher percentage of cholesterol (compare to other).

- function → Deliver cholesterol to cells throughout the body.

• Mostly LDL is taken up by liver for the synthesis of bile acid, because LDL contains cholesterol which used in the synthesis of bile acid & steroid hormones.



iv) HDL - High Density Lipoprotein →

- Produced in liver (70-80%) and also in small intestines.

• Mostly composed of phospholipids, cholesterol and proteins.

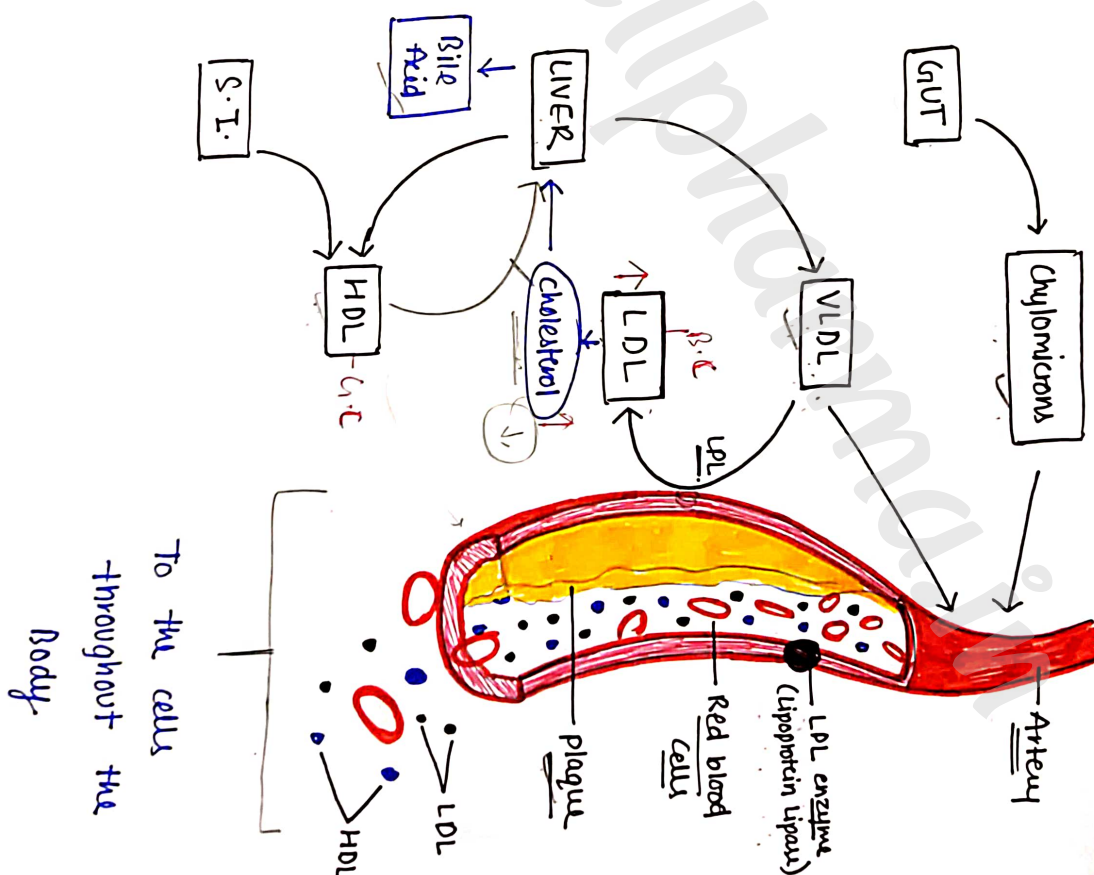
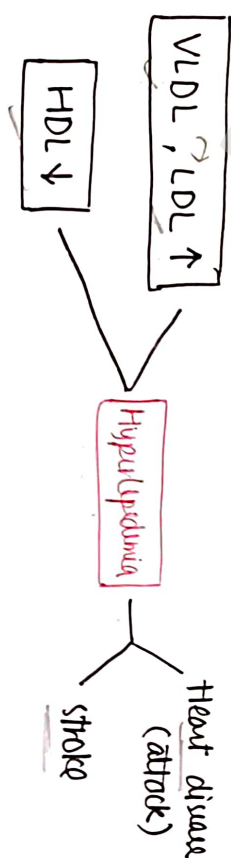
- functions → Transport excess cholesterol from the peripheral cells to the liver.

• It referred as a Good cholesterol and LDL referred as a Bad cholesterol.

Hypertipidemia →

- New hyperlipidemia occurs when LDL cholesterol level increases, because this excess cholesterol (LDL) accumulated in the innermost layer of the artery and leads to the formation of plaque (atheroma) which further cause heart disease.

- Due to this reason, LDL referred as Bad cholesterol.
- Now, HDL remove this excess cholesterol from peripheral so it referred as good cholesterol.



To the cells
throughout the
body.

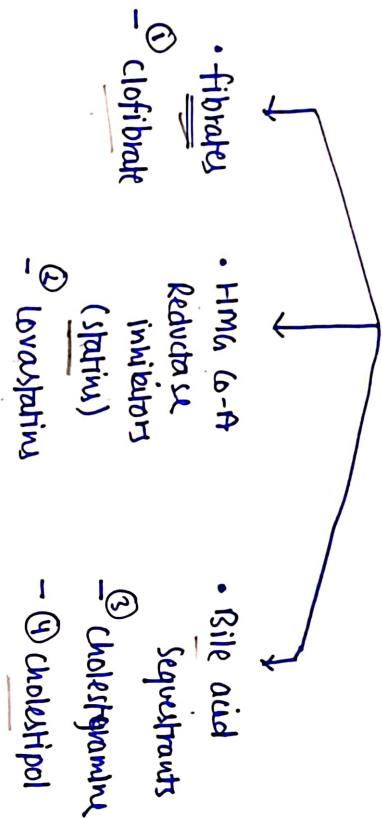
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• Classification (Acc. to N.C.-2 syllabus) -

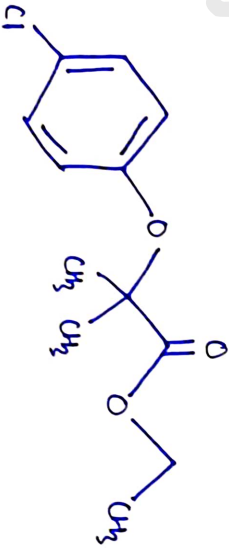


① clofibrate →

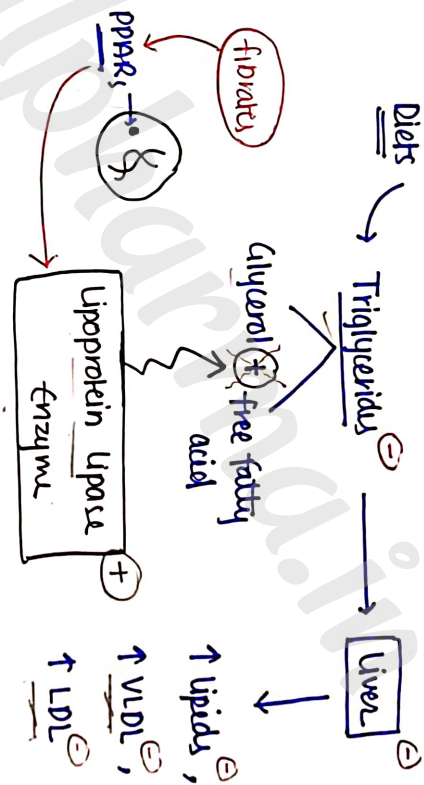
• It belongs to the class of fibrates which are the derivatives of fibric acid.

• It gives their action by decreasing triglyceride level & increases HDL levels.

Structure -



(MOA) -



- clofibrate binds and activates peroxisome proliferator activated receptor (PPARs) which are the number of nuclear receptor family that regulates lipid metabolism.
- Now, this PPARs increases the lipoprotein lipase enzymes which further degrades the triglycerides
- Results - decreasing production of hepatic VLDL, LDL and increasing HDL.

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uses -

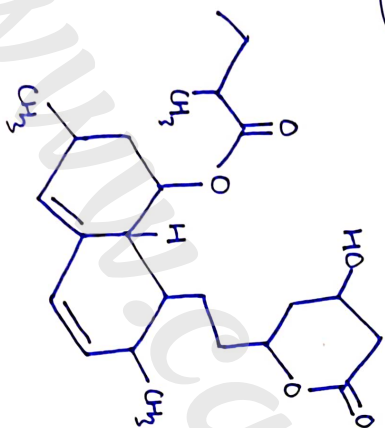
- used in the treatment of hypertriglyceridaemia.
- used for controlling the level of cholesterol and lipoproteins.

② lovastatin →

- It belongs to the class of

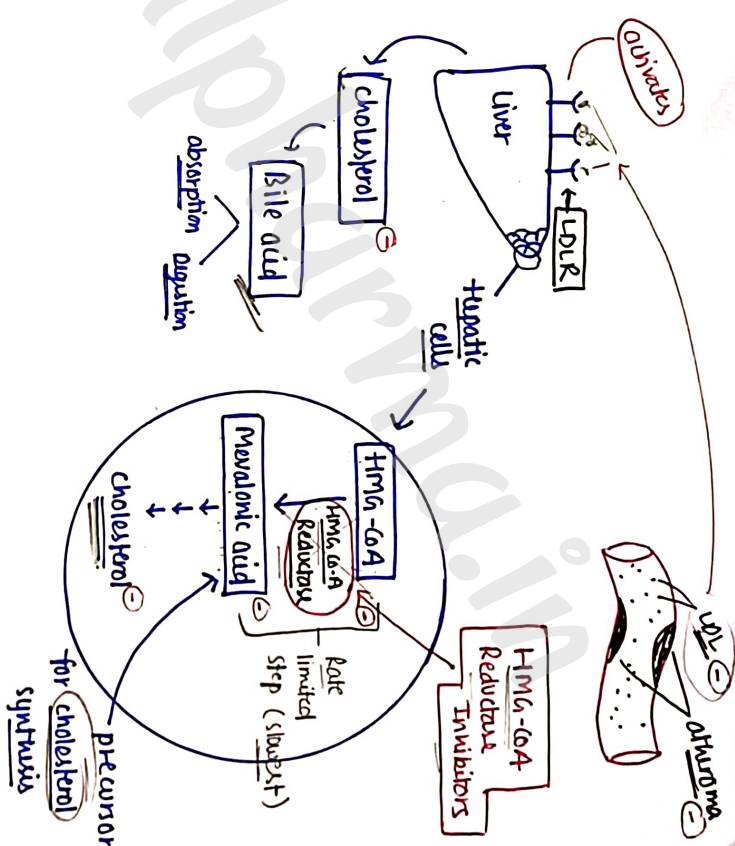
HMG - CoA Reductase Inhibitors. (statins).

Structure



MOA -

- lovastatin inhibits the enzyme HMG - CoA Reductase resulting in decrease in the production of cholesterol.



- Now, when liver starts decrease level of cholesterol it activates LDL (low density lipoprotein receptors)
- Now, LDL which present in blood is bind with LDL on liver and increase the level of cholesterol in liver.
- Due to this level of LDL is decrease in blood circulation which helps to treat atherosclerosis

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Uses -

- used to control high plasma levels of cholesterol & plasma lipids.
- prevent coronary artery disease
- also helps in prevention of cardiovascular diseases.

③

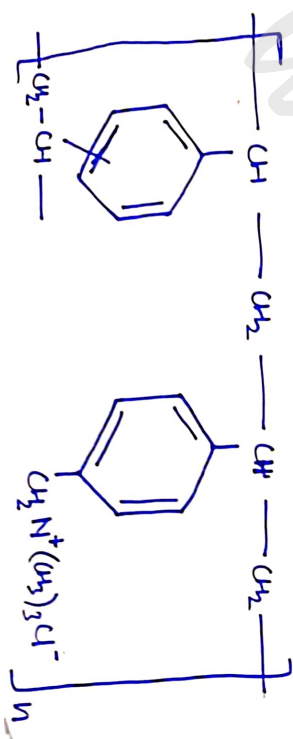
Cholestyramine →

- It belongs to the class of

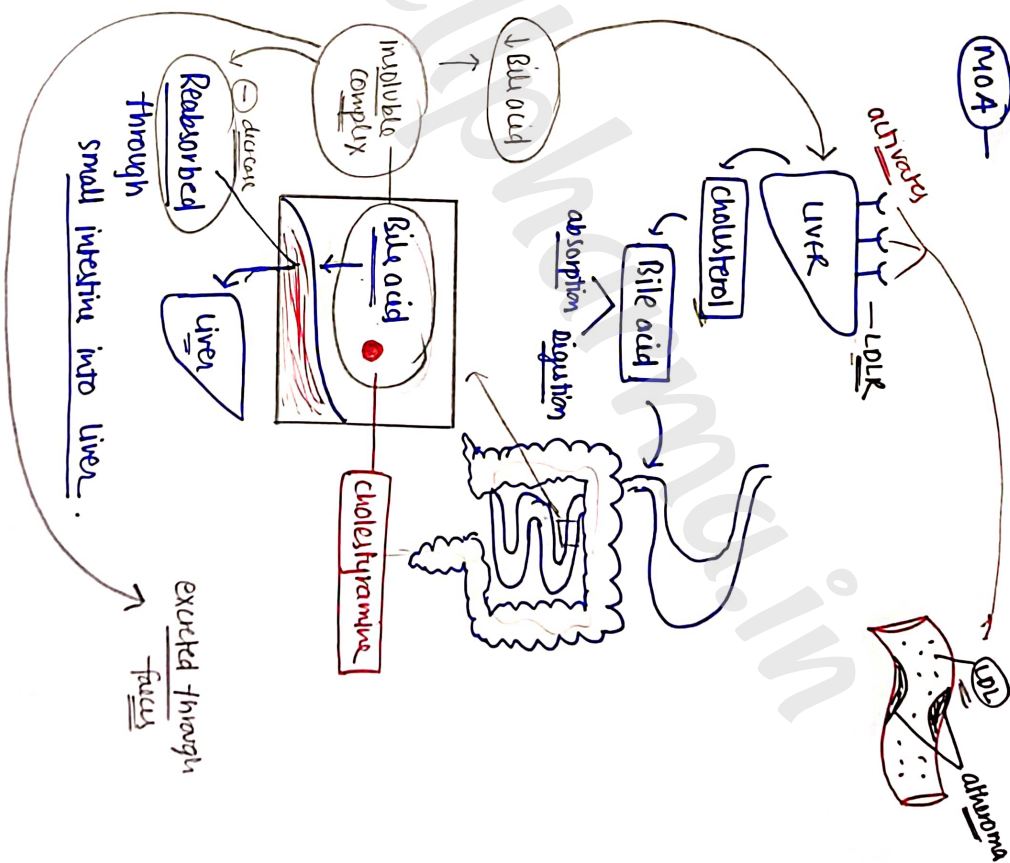
Bile acid sequestrants. (resins)

- these drugs give their action by lowering cholesterol in blood circulation.
- less effective than statins.

Structure -



Mechanism



- It binds with bile acid and its salts in the intestinal tract and cause their excretion
- Also inhibit Reabsorption of bile acid which

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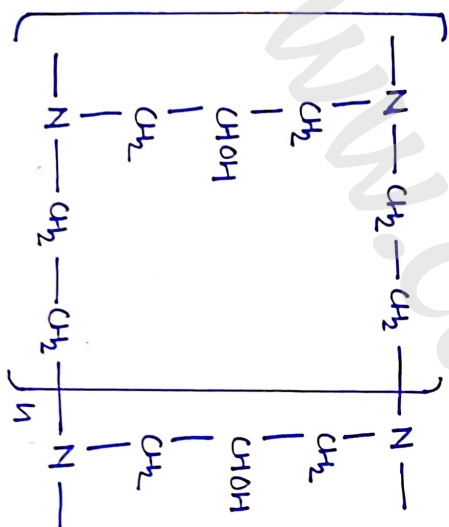
decrease level of bile acid in liver.

- Now, liver sees this and activates LDLR which further get the cholesterol rich LDL from blood circulation.

uses -

- used in reducing the elevated plasma cholesterol levels.
- also used in treatment of diarrhoea.
- used to treat hyperlipoproteinaemia.

④ colestipol →



uses + moA -

- same as ~~others~~ cholestyramine.

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COAGULANTS AND ANTI-COAGULANTS

Chapter - 3

- Coagulants → These are those substances or drugs which promote blood clotting
- Anti-coagulants → These are those substances which prevent blood clotting

• Blood clotting

- Haemostasis
 ↳ Haemo (connected to blood)
stasis (stop).

- It is sequence of responses that stop bleeding means reduces blood loss when blood vessels are damaged or ruptures.

③ mechanism of blood coagulation (clotting)

Hemostasis includes three steps that occurs in a rapid sequence:

- i) Vascular spasm → In this, the smooth muscles of a blood vessels contracts which

slow blood loss, due to release of activated platelets.



vasoconstriction → ↓ blood supply + loss

- ii) formation of platelet plug →

In this process, platelets aggregate and form plug which sticks to the damage blood vessels.

- iii) Blood coagulation / clotting →

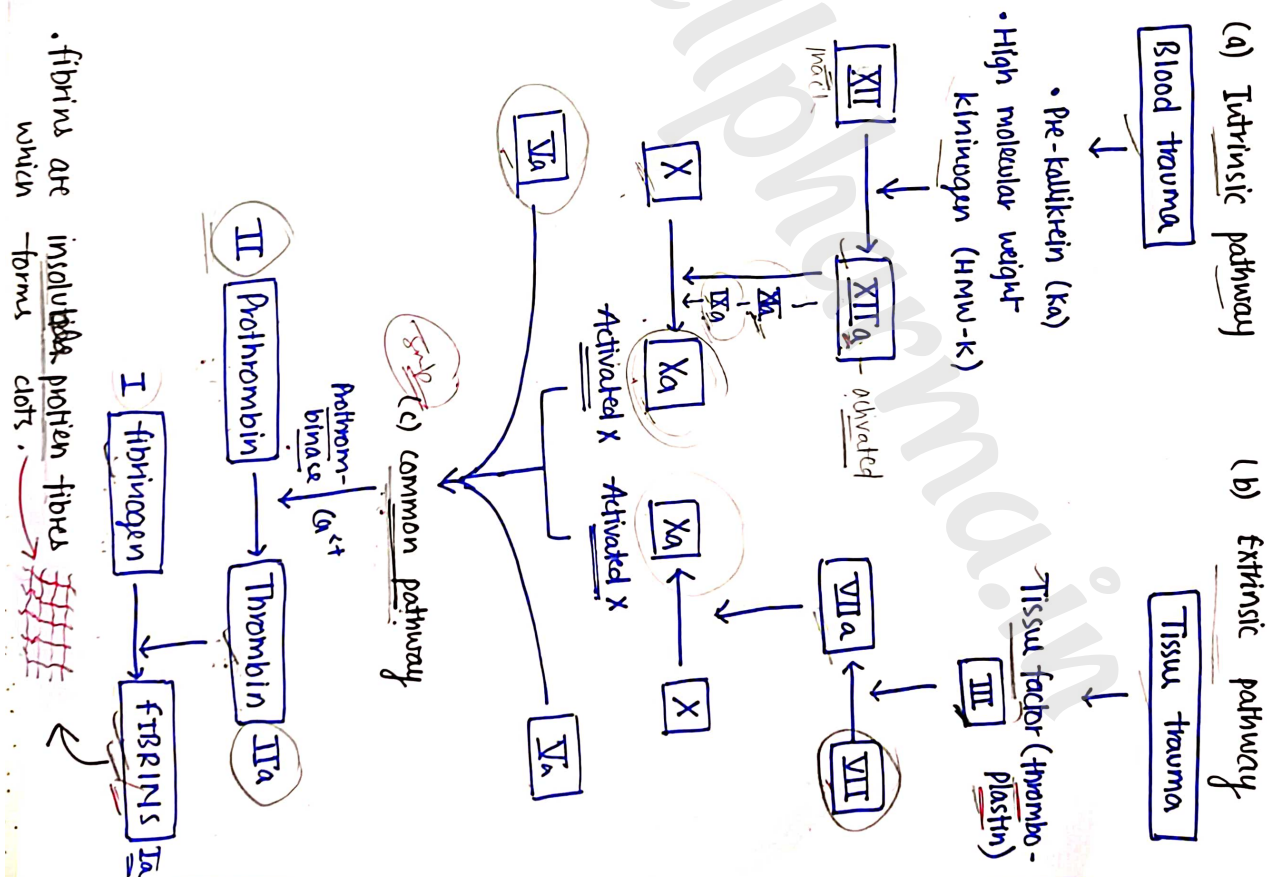
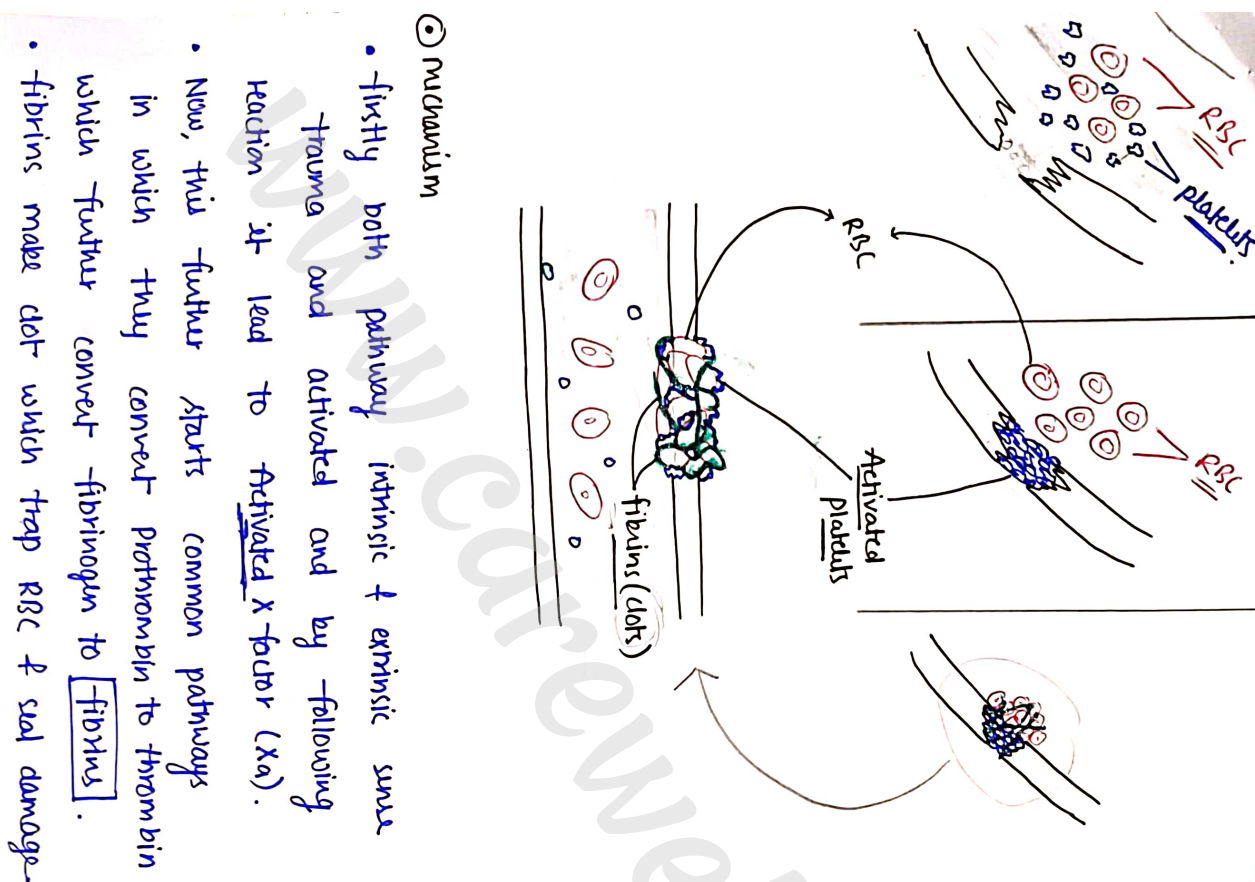
In this process, fibrins is formed which are insoluble proteins which further forms clots [network of fibrins].

Now, these clots trapped RBC and seal the damaged blood vessels.

- ③ Now, this process involves thirteen factors and three pathways.

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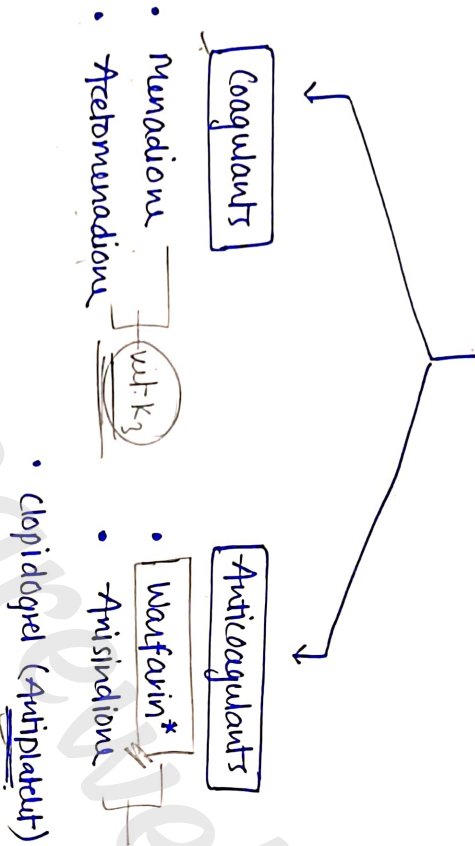
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CHAPTER - 3rd

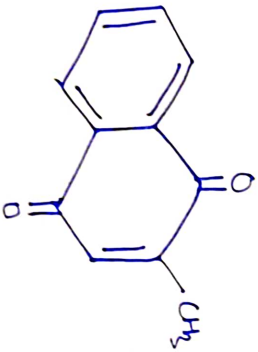
COAGULANTS & ANTICOAGULANTS



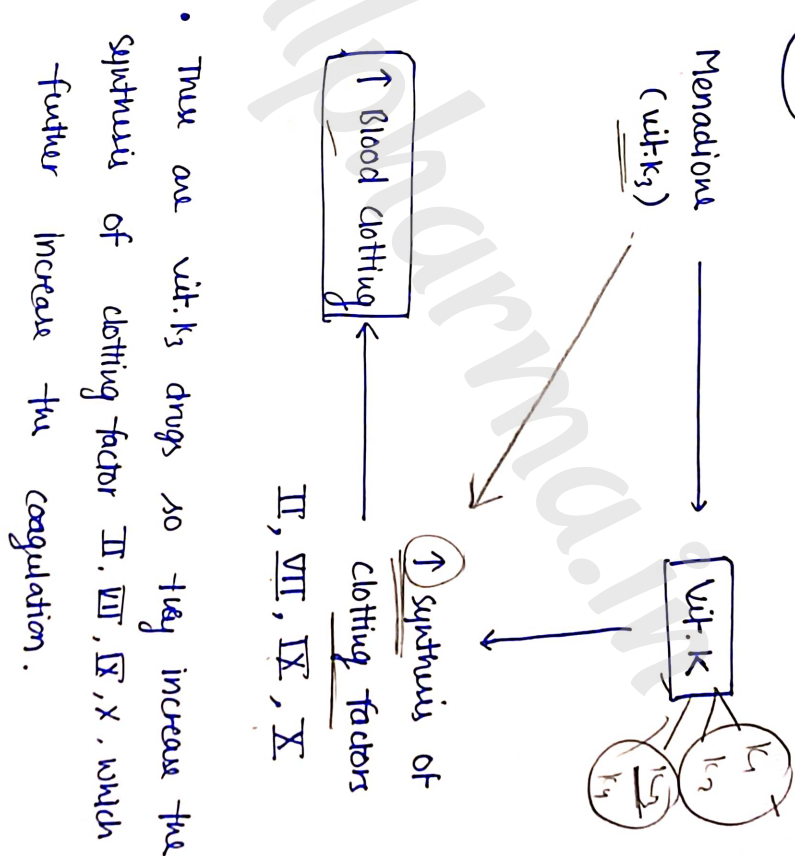
① Menadiolone →

It is an example of vitamin K₃ (synthetic).

Synthetic



MOA -



Uses

- Used in the treatment of bleeding
- increase coagulation
- also used in the deficiency of

vit. K.

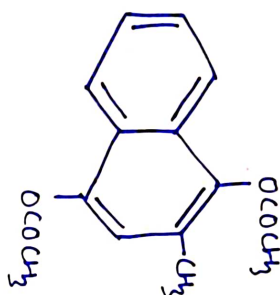
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② Acetaminouadione →

It is also an example of vit. K₃ which are synthetic analogues of vit. K.

Structure -



MOA -

• same as menadiione

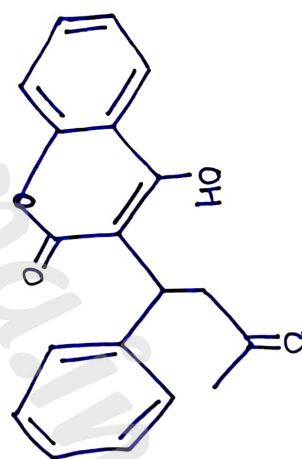
Uses -

• same as menadiione

③ Warfarin →

It belongs to the class of oral anticoagulants which are used to prevent blood coagulation.

Structure -



MOA -

Desarboxy
prothrombin

(factor - II, VII, IX, X)

Prothrombin

(factor - II, VII, IX, X)

↓ synthesis

hydroquinone

vit. K

vit. K
epoxide

NAD

NADH

Warfarin

↑ blocks

vit. K reductase

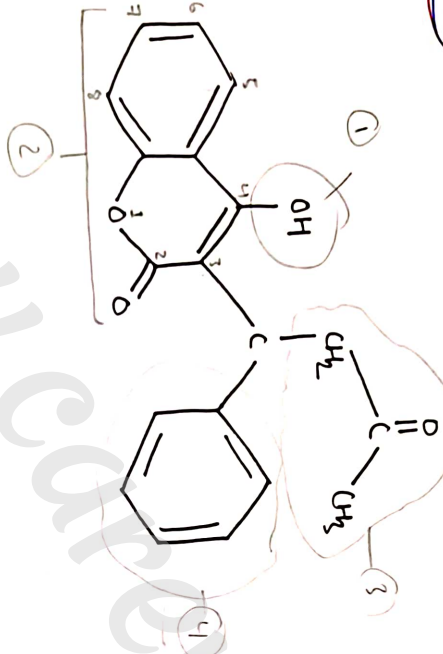
γ-glutamyl
carboxylase

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- It interfere with vit.k dependant synthesis of clotting factor (II, VII, IX, X) in liver by blocking the vit.k reductase.
- which further inhibit the blood coagulation.

SAR - STRUCTURE ACTIVITY RELATIONSHIP



- ① - OH group is essential for activity.
- SH group can be attached but it will reduce activity.
- ② - Coumarin ring is essential for activity.
- Position 3rd & 4th is responsible for different active derivatives.

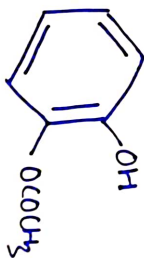
- ③ - ketone chain of 3 carbon produce potent compounds.

- it may be replaced with alkyl group but it will produce less potent compounds

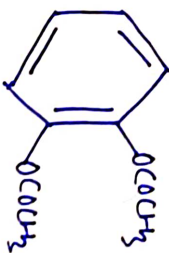
- ④ - Phenyl ring is essential for its activity

- Replacement with long alkyl group can increase the activity.

Synthesis

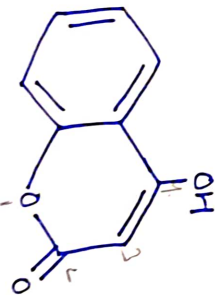


Acetylation



Methyl salicylate

Methylacetyl salicylate

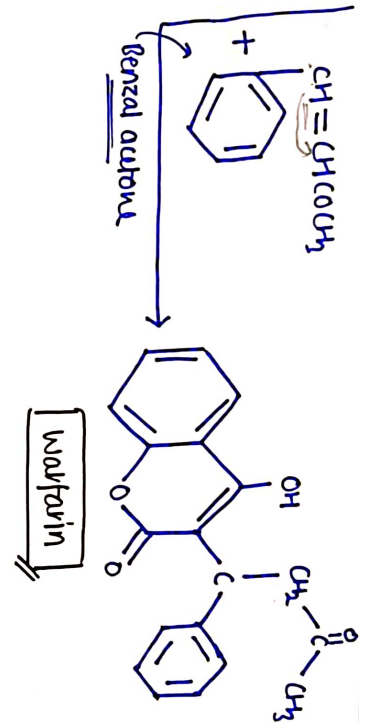


Cyclization

4-Hydroxycoumarin

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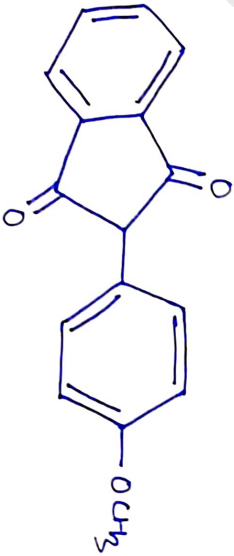
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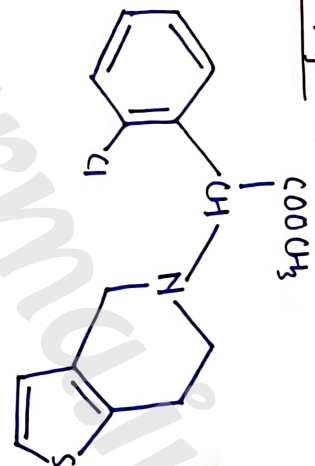
(Uses) -

- used in the treatment of deep vein thrombosis & pulmonary embolism.
- also used during surgical procedures to prevent thrombosis, to prevent stroke in atrial fibrillation.
- Reduce the risk of heart attack, stroke or death.

(4) Flavonoids → (same as warfarin) MOA/uses.

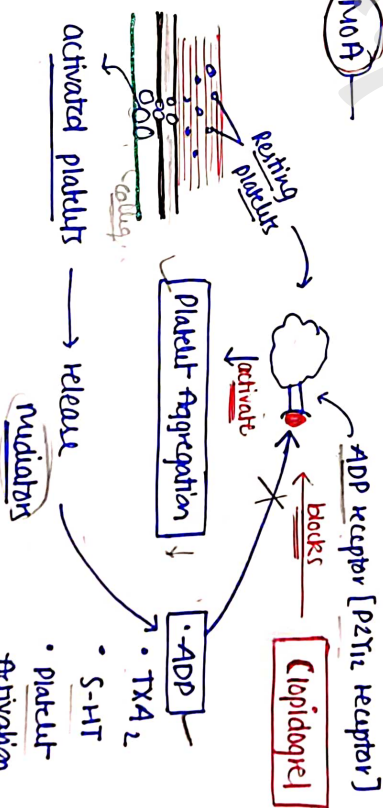


(5) Clopidogrel →



- It belongs to the class of anti-platelet drugs. (blood thinner).

(MOA)



- Clopidogrel inhibits the binding of ADP to its receptor on platelets.
- which inhibits the platelet activation & inhibit platelet aggregation.

(Uses) - as an antiplatelet medicine, prevent blood clots. ↓ heart attack, stroke.

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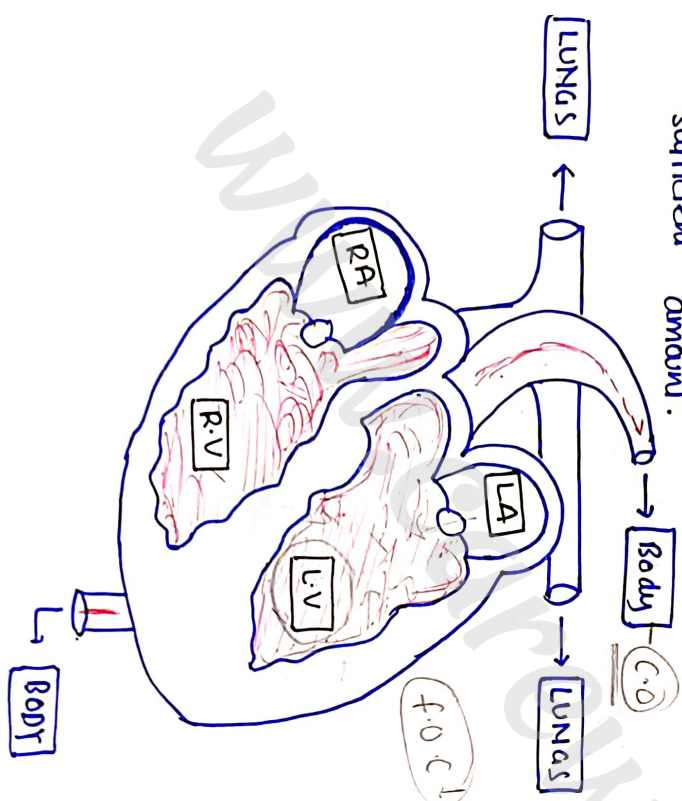
DRUG USED IN CONGESTIVE HEART FAILURE

Those drugs which are used in the treatment of congestive heart failure.

(CHF)

CONGESTIVE HEART FAILURE

It is a progressive condition in which the heart doesn't pump blood in a sufficient amount.



MANY REASONS

contractility of muscles ↓

force of contraction ↓

Blood Volume ↑ in heart
Cardiac output ↓

Congestive heart failure

• Heart attack

Decrease blood in systems + renal failure.
development of edema in lower extremities and the lungs

• Symptoms →

- Shortness of breath
- fatigue
- swollen legs
- Rapid heartbeat
- pain in the chest
- dizziness and many more...

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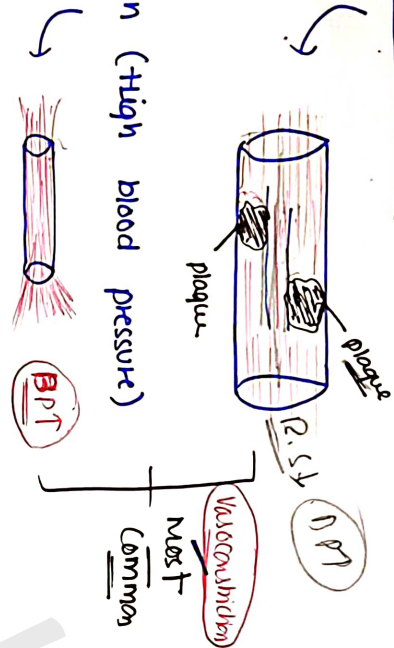
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Reasons →

• Coronary artery disease



• Hypertension (high blood pressure)

• Alcohol use (excess - abuse)

• Viral infection

• Thyroid disorder

• Heart rhythm abnormalities

Risk factors →

• High BP

• High cholesterol

• Diabetes

• Smoking, alcohol

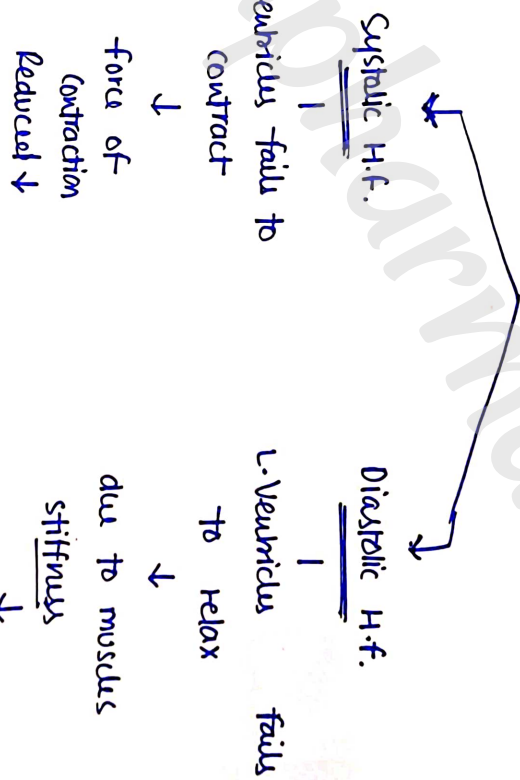
• Family history

• also heart valve disease

Types of CHF →

i) left sided CHF - most common type

- L. Ventricle does not pump blood properly.
- cause breathing difficulty.



ii) Right sided CHF -

Right ventricle does not pump properly into lungs. (Blood back up in blood vessels)

- cause fluid retention in lower extremities, abdomen and other vital organ.

• Sometimes passes from left to right.

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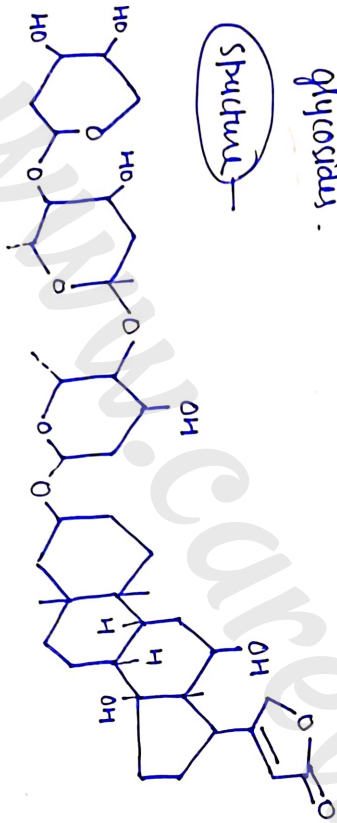
DRUGS

- Digoxin
 - Digitoxin
 - Mesigitide
 - Roseoxon
 - Tezosentan
- Cardiac glycosides (digitalis)
- M.I. + ischaemic*

① Digoxin →

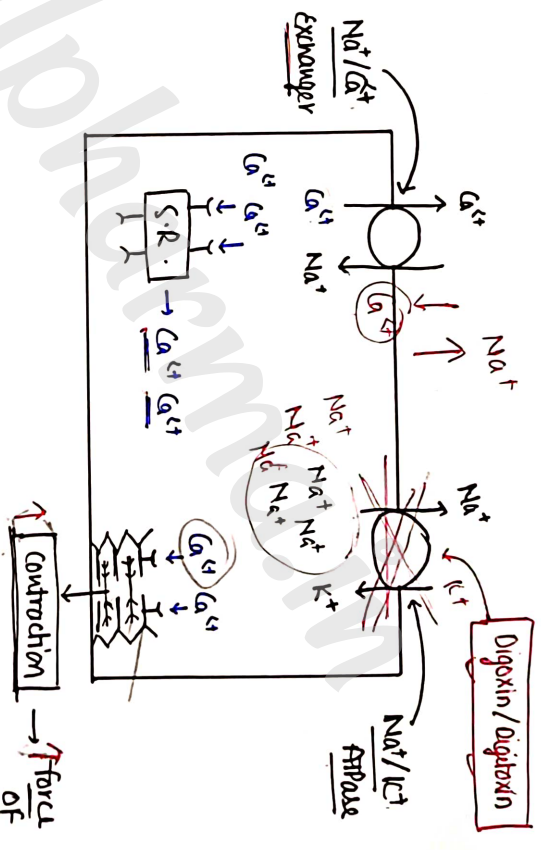
It belongs to the class of cardiac glycosides.

Structure



M.O.A

- Cardiac glycosides introduced which inhibit the Na^+/K^+ ATPase enzyme.



- Due to this, outflux of Na^+ is inhibited and accumulation of Na^+ inside the myocardial cell increases.
- Now, due to accumulation of Na^+ , $\text{Na}^+/\text{Ca}^{2+}$ exchanger change their way and it start to outflux the Na^+ & influx of Ca^{2+} .
- Now, Ca^{2+} ion accumulated intracellularly, which increase Ca^{2+} concentration which further increase contraction. (force of contraction).
- which helps in treatment of CHF.

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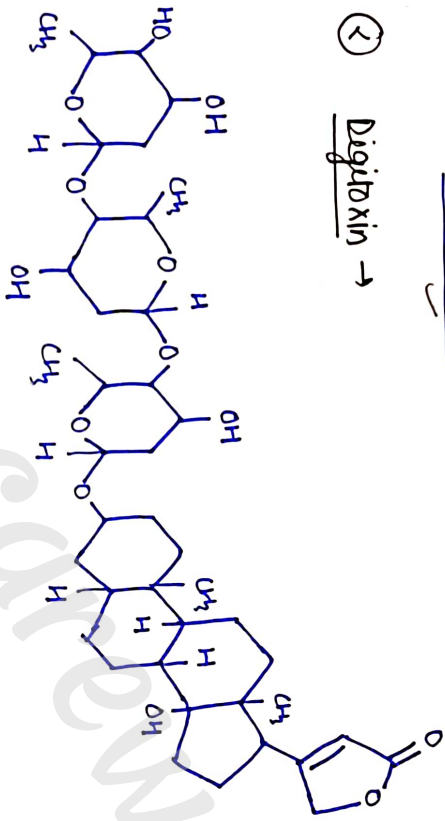
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uses -

- used in the treatment of congestive heart failure associated with atrial fibrillation.
- used to maintain normal heart rate in atrial arrhythmias.

② Digoxin →



MOA, uses

— same as Digoxin

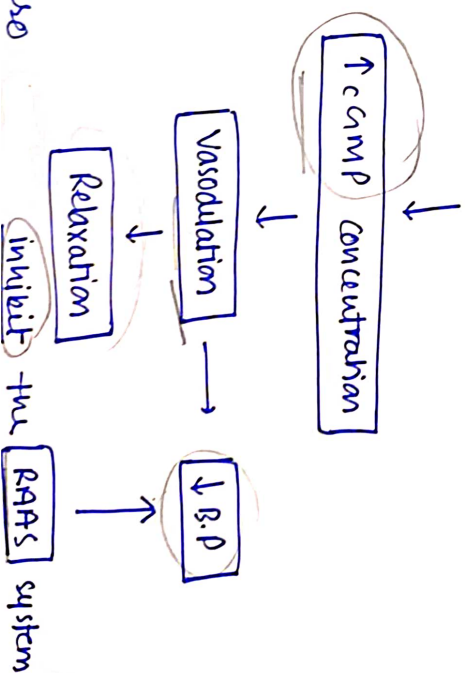
③ Nesiritide →

- It is one of the type of hormone which is secreted by cardiac muscle cells in the heart ventricles.

- It is a recombinant 32-amino acid human B-type natriuretic peptide (BNP) that have vasodilatory effects. ($\downarrow$ B.P.).

MOA -

- It binds with particulate guanylate cyclase receptor in vascular smooth muscles and endothelial cells



- It also

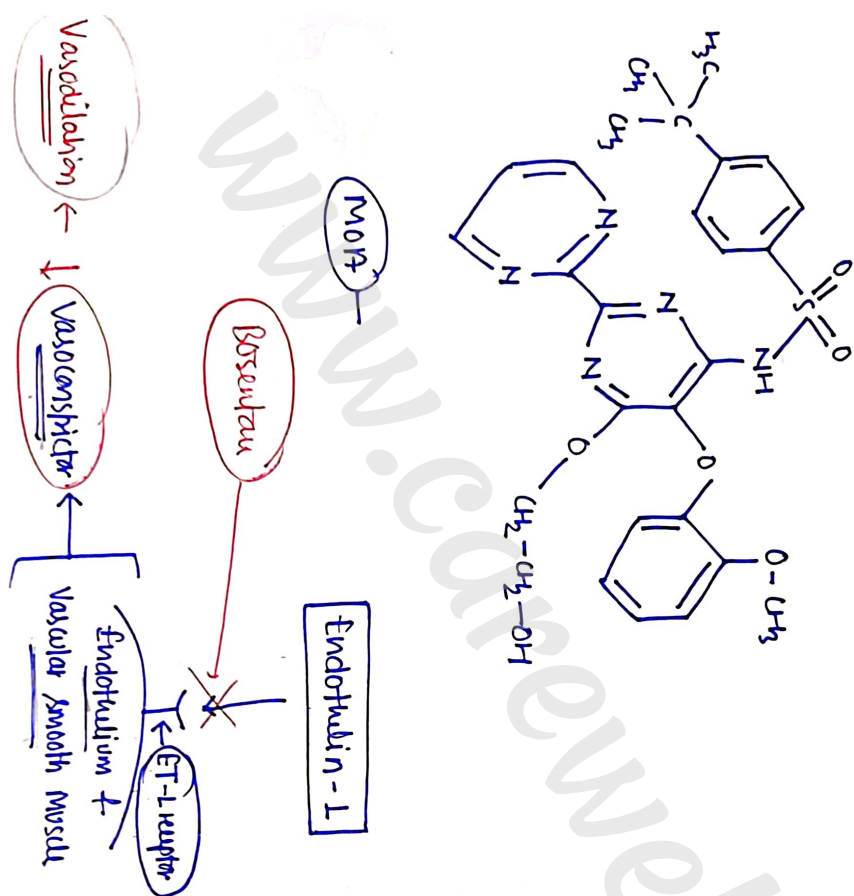
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uses -

- used in the treatment of CHF
- also help in vasodilation & diuresis.

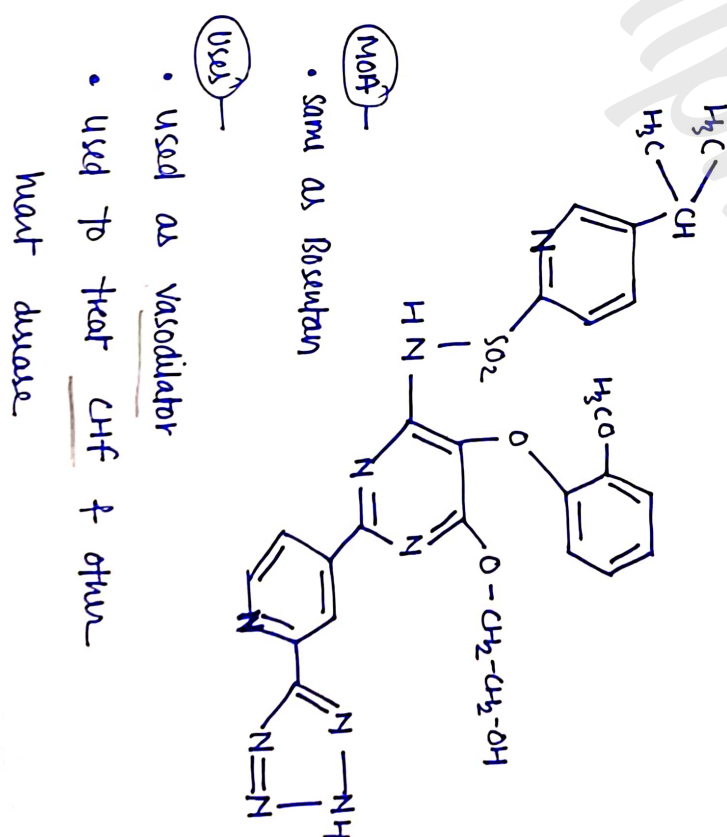
④ Bosentan →



uses -

- used in the treatment of pulmonary arterial hypertension (PAH)
- used in the treatment of systemic sclerosis or scleroderma.

⑤ Tezosentan →



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